Agar Plate Art & Bacterial Streaking

Three activities with agar plates: see bacteria on your own hands, grow a pure colony, and draw with bacteria

SAFETY FIRST — READ BEFORE YOU START

- Only use **premade, sealed** agar plates and approved lab strains. Never grow bacteria from toilets, wounds, the nose or mouth.
- After inoculating, **seal each plate with tape** (two pieces, not all the way around) and never open it again.
- For hand-print plates incubate at **room temperature (about 25–30 °C)**, not at 37 °C. 37 °C favours bacteria that live on people.
- Wash hands after the activity. No eating or drinking in the lab. Tie back long hair.
- **Ethanol is flammable:** never use it near an open flame.
- Disinfect all plates before throwing them away (see the end of this document).

Part A. Agar plate art: hand prints, before and after

AIM
To see the bacteria living on our fingers, and to test whether hand sanitiser reduces them.

Materials

- 2 premade nutrient agar plates per team
- marker pen (permanent)
- hand sanitiser (alcohol-based)
- sticky tape, gloves for those handling the plates after incubation (optional)

Procedure

1. Label the **bottom** of each plate with a marker: team name, date, and **"Before"** or **"After"**. Write on the edge so you do not block the view.
2. Place the plates on the table. Remove the lid of the **"Before"** plate.
3. Press **three fingers** firmly on the agar for about 5 seconds, then lift them. Do not poke through the agar.
4. Close the lid immediately.
5. Clean your hands thoroughly with sanitiser and wait until they are completely dry.
6. Remove the lid of the **"After"** plate. Press the **same three fingers** in the same way, and close the lid.
7. Seal both plates with tape. Turn them **upside down** (lid at the bottom) so that water drops do not fall on the agar.
8. Incubate at room temperature for **48–72 hours**.

9. Look (do not open!). Compare the two plates: count the colonies or estimate the number, and note the colours and shapes.

What to expect

You will usually see more colonies on the "Before" plate. But results vary: sanitiser reduces germs but does not remove all of them, and pressing harder or longer changes the result. Treat the experiment as a test, not a certainty. Ask students to predict the result before they see it.

Observation	Before plate	After plate
Number of colonies (count or estimate)		
Colours (white, yellow, orange…)		
Shapes and sizes of colonies		
Any fuzzy growth (mould)?		

Discuss

- Did the sanitiser kill everything? What could explain colonies on the "After" plate?
- Why did we use the *same* fingers, the same pressure and the same plate type each time? (A fair test.)
- Are all the bacteria on our skin harmful? (No. Most are harmless or helpful.)

Part B. Bacterial plate streaking

AIM

To separate bacteria so that each starts a single, isolated colony — the basis of a pure culture of E. coli (the harmless lab strain DH5α).

Materials

- A plate with *E. coli* DH5α colonies (the source plate) and fresh sterile agar plates
- inoculating loop (metal) with a **spirit lamp or Bunsen burner**, OR sterile pipette tips / toothpicks if no flame is available
- ethanol (kept well away from the flame), marker, tape

Procedure

1. Label the bottom of the new plate. Divide it into **four quadrants** with the marker.
2. **Sterilise the loop.** Heat it in the flame until it is red-hot, then let it cool for about 10 seconds (touch the cool agar edge to test). *Alternative:* use a sterile tip (autoclaved at 121 °C and 15 psi).
3. Touch the loop or tip to **one isolated colony** on the source plate.
4. **Quadrant 1.** Lift the lid just enough to work. Streak the loop gently back and forth over about a quarter of the plate. Do not dig into the agar.

5. **Re-sterilise** the loop in the flame and let it cool.
6. **Quadrant 2.** Turn the plate a quarter turn. Drag the cooled loop **once through the end of quadrant 1**, then zig-zag across the second quadrant. This drags out a few cells from the crowded area into fresh agar.
7. Re-sterilise the loop. Repeat for **quadrant 3** (pull from quadrant 2) and **quadrant 4** (pull from quadrant 3).
8. Flame the loop a last time. Close the plate, seal with tape, turn it upside down, and incubate at **37 °C for 24 hours** (or at room temperature for 48 hours).

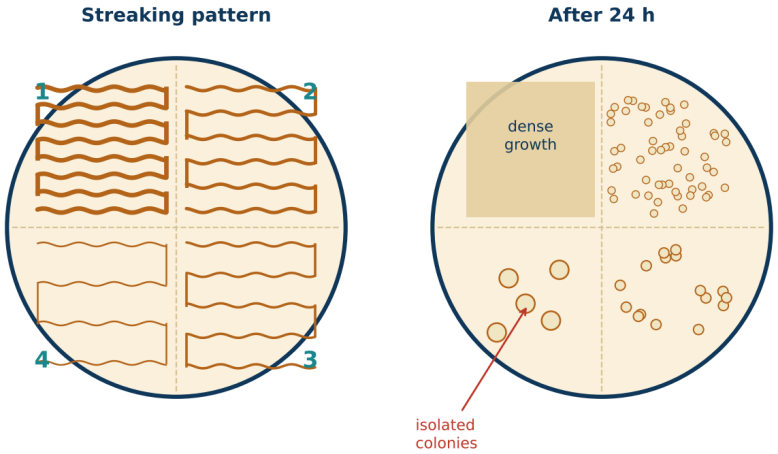


Figure 1. Left: the four-quadrant streaking pattern. Right: after incubation, quadrant 1 has dense growth while quadrant 4 shows separate single colonies. *Source: Own illustration*

TROUBLESHOOTING

- *Agar torn:* the loop was too hot or pressed too hard — wait to cool and use light pressure.
- *One solid layer everywhere:* the loop was not re-sterilised, or you streaked over the same area too many times.
- *No growth:* the loop was too hot and killed the cells, or the plate was incubated too cold.

Part C. Draw with bacteria (optional)

Once students know streaking, they can create a picture on an agar plate. Dip a sterile loop or cotton bud in a harmless culture (such as the DH5α used above) and draw a simple design — a smiley face, a leaf, initials — on a fresh plate. Seal, invert and incubate for 24–48 hours. The design appears as cream-coloured lines. Teams can compare art, and discuss why smooth, even lines need a light touch.

COLOURFUL IDEAS

Some labs and iGEM teams use harmless engineered bacteria that make coloured or fluorescent proteins for plate art. Use these only with an approved strain and under the teacher's supervision.

Cleaning up

- Keep all plates sealed. Before disposal, either **autoclave** them (121 °C, 15 psi, at least 15–20 minutes) or cover with fresh **10 % household bleach** and leave for at least 30 minutes.
- Wipe benches with disinfectant or 70 % ethanol before and after work.
- Wash hands with soap and water.

Check your understanding

1. Why do we turn plates upside down before incubating?
2. Why do we re-sterilise the loop between quadrants?
3. In which quadrant do you expect single colonies, and why?
4. Why is it not safe to open plates grown from hand prints?